

**UNIVERSITY OF GHANA**  
**DEPARTMENT OF COMPUTER SCIENCE**  
**DCIT 208 – Software Engineering**  
Individual Assignment 3: Use-Case-to-Story Traceability Card

|                              |                                                                     |
|------------------------------|---------------------------------------------------------------------|
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| <b>Student ID</b>            | 22337992                                                            |
| <b>Team name</b>             | TECHCARE                                                            |
| <b>Project title</b>         | Pharmacy Web Application                                            |
| <b>Client/project domain</b> | Community pharmacy operations – prescription verification and sales |
| <b>Date submitted</b>        | 13 July 2026                                                        |

## 1. Selected use case

**Use case name:** Customer uploads a prescription receipt for verification

**Source evidence:**

Why this use case matters to the client: Prescription medicines cannot legally be sold without a valid prescription. Requiring the customer to upload a receipt or prescription image before an order is confirmed lets the pharmacist verify authenticity, reduces the risk of dispensing controlled or prescription-only drugs without authorization, and creates an auditable record the pharmacy can refer back to.

## 2. Use case card

|                      |                                                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>         | Customer (registered web user placing a medication order)                                                                          |
| <b>Goal</b>          | Upload a photo or scanned copy of a valid prescription so the pharmacist can verify it before the order is approved                |
| <b>Trigger</b>       | Customer selects prescription-only medication and proceeds to checkout, or opens the 'Upload Prescription' option from their order |
| <b>Preconditions</b> | Customer is logged in; an order containing at least one prescription-required item exists; upload service/storage is reachable     |

### Main success flow

1. Customer opens the order and selects 'Upload Prescription Receipt'.
2. Customer selects an image or PDF file of the prescription from their device.
3. System validates the file (type, size, that a file was selected).
4. System stores the file and links it to the order with status 'Pending Verification'.
5. System notifies the pharmacist/admin that a new prescription is awaiting review.
6. Customer sees an on-screen confirmation that the receipt was received and is under review.

### Alternate flow

Customer uploads more than one page (e.g., front and back of a physical prescription). System accepts multiple files for the same order and groups them under one verification record.

### Exception/failure flow

Customer selects a file that is not an image or PDF, or exceeds the maximum size limit. System rejects the upload, displays a specific error message ('Only JPG, PNG or PDF files under 5MB are accepted'), and does not create an order-linked record.

**Postconditions:** A prescription record exists with status 'Pending Verification' linked to the order; the pharmacist queue shows the new item; the customer's order remains on hold until verification is completed

### 3. User story

*As a customer, I want to upload a photo or scan of my prescription during checkout, so that the pharmacist can verify it and approve my order for a prescription-only medication.*

### 4. Acceptance criteria

#### AC1 – Successful path

Given a logged-in customer has an order containing a prescription-required item and selects a valid image or PDF file under the size limit, when the customer submits the upload, then the system stores the file, links it to the order, sets the status to 'Pending Verification', and displays a confirmation message with a reference number.

#### AC2 – Error and exception path

Given a logged-in customer attempts to upload a file that is not an accepted type (e.g., .exe, .docx) or exceeds the size limit, when the customer submits the upload, then the system rejects the file, displays a specific validation error naming the accepted formats and size limit, and does not create any order-linked record.

### 5. Non-functional requirement link

**NFR category:** Security & Privacy

**NFR statement:** Uploaded prescription images, which contain personal health information, must be stored so that only the uploading customer and authorized pharmacy staff (admins) can access them, and access must be logged.

**Why it matters:** A prescription receipt reveals what medical condition a customer may have and what medication they take. If any authenticated user could view another customer's upload by guessing a file URL, or if files were stored without access control, the system would expose sensitive health data and could breach data protection expectations for a pharmacy platform. Restricting access and logging who viewed a file protects customer trust and gives the pharmacy an audit trail if a dispute arises.

### 6. GitHub issue link

**Issue title:** Prescription Receipt Upload for Verification

**Issue URL:** <https://github.com/Dept-of-Comp-Sci-University-of-Ghana/dcit208-client-project-2026-team-engineering-repository-seg26-23techcare>

### 7. Test idea

**Test name:** Reject invalid file type on prescription upload

**Input :** Logged-in customer with a pending prescription-required order attempts to upload a renamed executable file (test.exe renamed to test.jpg) as the prescription receipt

**Expected result:** Upload is rejected based on actual file content (not just the extension), a clear error is shown, and no prescription record is created in the database for that order

### 8. Request-flow thinking

|                  |                                                                                   |
|------------------|-----------------------------------------------------------------------------------|
| Device/interface | Web browser (desktop or mobile), using the pharmacy web app's checkout/order page |
|------------------|-----------------------------------------------------------------------------------|

|                                          |                                                                                                                                                                                                                                                          |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Route/screen/endpoint</b>             | POST /api/orders/order_id/prescription-upload (Flask route), called from the 'Upload Prescription' screen                                                                                                                                                |
| <b>Data sent</b>                         | Multipart form data: the file itself, order ID, and the authenticated customer's session/token                                                                                                                                                           |
| <b>Validation point</b>                  | Server-side validation in the Flask route: file present, MIME type/content check (not just extension), file size limit, order belongs to the requesting customer                                                                                         |
| <b>Database/external service touched</b> | SQLite: prescriptions table (order_id, file_path, status, uploaded_at, verified_by); file itself saved to server storage or an object storage folder referenced by file_path                                                                             |
| <b>Response</b>                          | JSON response with success flag, a reference/tracking number, and the new status ('Pending Verification'), or an error object with a message code                                                                                                        |
| <b>Failure point</b>                     | If the file fails validation, if storage/disk write fails, or if the database insert fails, the request returns an error and no partial record is left (transaction rolled back)                                                                         |
| <b>Security/privacy concern</b>          | Prescription images contain personal health data; the endpoint must confirm the order belongs to the requesting customer, files must not be publicly listable, and only authorized pharmacist/admin roles should be able to open a given customer's file |

## 9. AI disclosure

**AI used?:** Yes

**Tool used:** Claude (Anthropic)

What it helped with: Structuring the use-case card into the required sections, drafting initial wording for the user story, acceptance criteria, NFR justification, and request-flow answers based on the Prescription Receipt Upload feature I was individually assigned on the team project.

## 10. Integrity declaration

I understand the use case, story, acceptance criteria and traceability links submitted here, and I can explain them orally if called upon.

**Signature :** AIDOO ROBERT KWABENA

## Traceability row (Part F)

| Requirement/Source                                                  | Use Case                                                               | User Story                                                                                                    | NFR                | GitHub Issue                             | Acceptance Criteria                                                                                                             | Test Idea                                                                            | Status      |
|---------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------|
| Team backlog item PH-07 (client interview note, D3 sprint planning) | Customer uploads a prescription receipt for verification before pickup | As a customer, I want to upload a photo/scan of my prescription so that the pharmacist can verify it before I | Security & Privacy | [team-repo]/issues/[#] (see Part F note) | AC1: valid file + fields -> stored, status 'Pending Verification'.<br>AC2: invalid file type/size -> rejected with clear error, | Upload a .exe file renamed as .jpg; expect rejection and no database record created. | In Progress |

| Requirement/Source | Use Case | User Story             | NFR | GitHub Issue | Acceptance Criteria | Test Idea | Status |
|--------------------|----------|------------------------|-----|--------------|---------------------|-----------|--------|
|                    |          | collect my medication. |     |              | nothing stored.     |           |        |